Heart of Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Slope and equations of lines**

Find the slope of the line through $(2, 5)$ and $(6, 13)$.

$m = \frac{13-5}{6-2} = \frac{8}{4} = 2$.

2 **[Foundational]** • **Target: Slope and equations of lines**

Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.

$y = -3x + 7$.

3 **[Foundational]** • **Target: Slope and equations of lines**

Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.

A perpendicular slope is the negative reciprocal of $\frac{2}{5}$, which is $-\frac{5}{2}$.

4 **[Foundational]** • **Target: Systems of equations by substitution**

Solve the system: $y = 3x - 4$ and $2x + y = 11$.

Substitute: $2x + (3x - 4) = 11$, so $5x - 4 = 11$, giving $5x = 15$, $x = 3$. Then $y = 3(3) - 4 = 5$. Solution: $(3, 5)$.

5 **[Foundational]** • **Target: Systems of equations by elimination**

Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.

Adding the two equations eliminates y: $4x = 20$, so $x = 5$. Substituting into $x - 2y = 1$: $5 - 2y = 1$, so $2y = 4$, giving $y = 2$. Solution: $(5, 2)$.

6 **[Foundational]** • **Target: Interpreting linear models**

A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.

The 20 represents a fixed monthly fee (the cost when $m = 0$ minutes are used), and the 0.05 represents the cost per minute — each additional minute of calls adds 0.05 dollars to the bill.

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7 [Foundational] • Target: Interpreting linear models

Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d-mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

Set equal: $30 + 0.2d = 0.35d$, so $30 = 0.15d$, giving $d = 200$ miles. At 150 miles: Company A charges $30 + 0.2(150) = 60$ dollars; Company B charges $0.35(150) = 52.5$ dollars. Company B is cheaper at 150 miles.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Systems with no solution or infinitely many solutions

For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

For infinitely many solutions the equation must hold for every x, so the coefficients of x must match: $k = 6$ (the constants already agree, $4 = 4$).

9 [Intermediate] • Target: Systems with no solution or infinitely many solutions

Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

Multiplying the first equation by 2 gives $4x - 6y = 12$. This contradicts the second equation, $4x - 6y = 10$ — the same left-hand expression cannot equal two different values, so the lines are parallel and never intersect.

10 [Intermediate] • Target: Compound inequalities

Solve the compound inequality $-3 < 2x + 1 \leq 9$.

Subtract 1 from all three parts: $-4 < 2x \leq 8$. Divide by 2: $-2 < x \leq 4$.

11 [Intermediate] • Target: Multi-step word problems

Two numbers have a sum of 50 and a difference of 12. Find the two numbers.

Let the numbers be x and y with $x + y = 50$ and $x - y = 12$. Adding: $2x = 62$, so $x = 31$. Then $y = 50 - 31 = 19$. The numbers are 31 and 19.

12 [Intermediate] • Target: Multi-step word problems

The sum of three times a number and 7 equals 5 less than twice the number. Find the number.

$3x + 7 = 2x - 5$. Subtract $2x$: $x + 7 = -5$, so $x = -12$.

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13 [Intermediate] • Target: A well-designed word problem: ticket sales at a community theater

This question has three parts. A community theater sells adult tickets for 15 dollars and child tickets for 8 dollars. On Saturday the theater sold 120 tickets total and collected 1,450 dollars. (a) Set up a system of two equations for the number of adult tickets a and child tickets c sold. (b) Solve the system to find how many of each ticket type were sold. (c) On Sunday the theater raises the adult price to 18 dollars but keeps the child price the same. If the same number of adult and child tickets sell as on Saturday, find the new total revenue.

(a) $a + c = 120$ and $15a + 8c = 1450$. (b) From the first equation, $a = 120 - c$. Substitute: $15(120 - c) + 8c = 1450$, so $1800 - 15c + 8c = 1450$, giving $1800 - 7c = 1450$, so $7c = 350$, $c = 50$. Then $a = 120 - 50 = 70$. So 70 adult tickets and 50 child tickets were sold. (c) New revenue $= 18(70) + 8(50) = 1260 + 400 = 1660$ dollars.

14 [Intermediate] • Target: Slope and equations of lines

For what value of the constant a will the system of equations $ax + 4y = 7$ and $3x + 2y = 5$ have no solution?

Step 1: A linear system has no solution when the lines are parallel and distinct (slopes are equal, y-intercepts differ). In standard form, this requires: $\frac{a}{3} = \frac{4}{2}eq\frac{7}{5}$ Step 2: Solve the slope equality: $\frac{a}{3} = 2 \Rightarrow a = 2 \times 3 = \mathbf{6}$ Check: When $a = 6$, the ratio of coefficients is $\frac{6}{3} = \frac{4}{2} = 2$, while the constant ratio is $\frac{7}{5} = 1.4eq2$. Therefore, the lines are strictly parallel with no solution.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Systems of equations by substitution • Cross-Lesson: 2.01 Quadratics & Polynomials

A school theater sold 250 tickets for a play, raising a total of $1,750$. $Adulttickets cost9$ each and student tickets cost 5 each. How many adult tickets were sold?

Set up system of equations: Let a be the number of adult tickets and s be student tickets: $\{a + s = 250; 9a + 5s = 1750$ Elimination method: Multiply the first equation by 5: $5a + 5s = 1250$. Subtract from the second equation: $(9a + 5s) - (5a + 5s) = 1750 - 1250 \Rightarrow 4a = 500 \Rightarrow \mathbf{a = 125}$. Exactly 125 adult tickets were sold (and $s = 250 - 125 = 125$ student tickets).

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16 [Challenging] • Target: Systems of equations by elimination • Cross-Lesson: 3.01 Functions & Graphs

Level 4 · SynthesisIn the system of equations $ax + 8y = 24$ and $6x + by = 8$, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

The constant ratio from the first equation to the second is $\frac{24}{8} = 3$. For infinitely many solutions, every coefficient in the first equation must be 3 times the corresponding coefficient in the second: $a = 3 \times 6 = 18, \quad 8 = 3b \Rightarrow b = \frac{8}{3}$. $a + b = 18 + \frac{8}{3} = \frac{62}{3}$.

17 [Challenging] • Target: Absolute value equations • Cross-Lesson: 4.01 Ratios, Rates & Data

Find all real solutions to the equation $|2x - 7| = 3x - 1$.

Step 1: Split into two cases based on the definition of absolute value: Case 1: $2x - 7 = 3x - 1 \Rightarrow -7 + 1 = 3x - 2x \Rightarrow x = -6$. Check Case 1: $|2(-6) - 7| = |-19| = 19$, but the right side is $3(-6) - 1 = -19$. Since $19eq -19$, $x = -6$ is an extraneous solution. Case 2: $2x - 7 = -(3x - 1) = -3x + 1 \Rightarrow 2x + 3x = 1 + 7 \Rightarrow 5x = 8 \Rightarrow x = \frac{8}{5} = 1.6$. Check Case 2: $|2(1.6) - 7| = |3.2 - 7| = |-3.8| = 3.8$. Right side: $3(1.6) - 1 = 4.8 - 1 = 3.8$ (Valid ✓). The o.

18 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 5.01 Statistics & Probability

Solve the linear system $3x + 2y = 16$ and $x - y = 2$ for x and y.

Step 1: From the second equation, isolate x: $x = y + 2$ Step 2: Substitute into the first equation: $3(y + 2) + 2y = 16 \Rightarrow 3y + 6 + 2y = 16 \Rightarrow 5y + 6 = 16 \Rightarrow 5y = 10 \Rightarrow \mathbf{y = 2}$ Step 3: Substitute back to find x: $x = 2 + 2 = \mathbf{4}$ The unique solution is **(4, 2)**.

19 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 6.01 Geometry

Level 2 · MediumA theater sold a total of 250 adult and student tickets for an evening performance, collecting $2,400. If adult tickets cost 12 each and student tickets cost 8 each, how many student tickets were sold?

Let a be adult tickets and s be student tickets: $\{a + s = 250; 12a + 8s = 2400$ Multiply the first equation by 12: $12a + 12s = 3000$. Subtract the second equation: $(12a + 12s) - (12a + 8s) = 3000 - 2400 \Rightarrow 4s = 600 \Rightarrow s = 150$. The theater sold 150 student tickets.

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20 [Challenging] • Target: Compound inequalities • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

Optimal Strategy: On the Digital SAT's Item Response Theory (IRT) scoring model, fully correct answers yield substantially higher raw theta weights than rushed guesses. Allocating 3.5 minutes each to 2 questions using Desmos graphing/regression guarantees 2 correct hard items (+40–60 scaled points), while placing a rapid educated guess on the 3rd question preserves time with zero guessing penalty.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving linear equations

Two numbers have a sum of 63 and a difference of 9. Find the larger number.

a. 38 **b.** 36 ✓

c. 34 **d.** 27

Let the numbers be x and y with $x + y = 63$ and $x - y = 9$. Adding: $2x = 72$, so $x = 36$. Then $y = 63 - 36 = 27$. The larger number is 36.

22 [Foundational] • Target: Linear inequalities

For what value of constant k will the system of linear equations $3x - 5y = 8$ and $6x + ky = 16$ have infinitely many solutions?

a. -10 ✓ **b.** 10

c. -5 **d.** 6

For infinitely many solutions, coefficients must be proportional: $\frac{3}{6} = \frac{-5}{k} = \frac{8}{16} \Rightarrow \frac{1}{2} = \frac{-5}{k} \Rightarrow k = -10$.

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23 [Foundational] • Target: Slope and equations of lines

In the system of equations below, c is a constant: $3x - 6y, = 9; x - 2y, = c$ If the system has infinitely many solutions, what is the value of c?

a. 3 ✓

b. 9

c. -3

d. 1

For a linear system to have infinitely many solutions, the two equations must represent the identical line. Dividing the first equation by 3 gives $x - 2y = 3$. Comparing with the second equation $x - 2y = c$ yields $c = 3$.

24 [Foundational] • Target: Systems of equations by substitution

A system of two linear equations has no solution. Which of the following must be true about the two lines?

a. **They have the same slope but different y-intercepts** ✓

b. They have different slopes

c. They have the same slope and the same y-intercept

d. They are perpendicular

A system with no solution corresponds to two parallel, distinct lines: equal slopes, different y-intercepts.

25 [Foundational] • Target: Systems of equations by elimination

In the system of equations below, k is a constant: $- - 4x + y, = 1; kx - 3y, = -15$ For which value of k does the system have no solution?

a. -4

b. 12

c. 3

d. -12 ✓

Two linear equations have no solution if and only if their slopes are equal and their y-intercepts differ (representing distinct parallel lines). Rewriting each equation in slope-intercept form $y = mx + b$: From Equation 1: $y = -4x + 1$, slope $m_1 = -4$. From Equation 2: $-3y = -kx - 15 \Rightarrow y = \frac{k}{3}x + 5$, slope $m_2 = \frac{k}{3}$. Setting slopes equal: $\frac{k}{3} = -4 \Rightarrow k = -12$.

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26 [Foundational] • Target: Absolute value equations

A theater sells adult tickets for $x14andchildticketsfor\10$. On a night when 12 tickets were sold for a total of $x148$, how many adult tickets were sold?

a. **7 ✓**

b. 5

c. 6

d. 8

$x + y = 12,\ 14x + 10y = 148 \Rightarrow x = 7,\ y = 5$

27 [Foundational] • Target: Interpreting linear models

A theater sells large tickets for $x28andsmallticketsfor\9$. On a night when 16 tickets were sold for a total of $x201$, how many large tickets were sold?

a. 13

b. **3 ✓**

c. 4

d. 2

$x + y = 16,\ 28x + 9y = 201 \Rightarrow x = 3,\ y = 13$

28 [Foundational] • Target: Systems with no solution or infinitely many solutions

A theater sells premium tickets for $x8andstandardticketsfor\3$. On a night when 8 tickets were sold for a total of $x39$, how many premium tickets were sold?

a. 2

b. 5

c. 4

d. **3 ✓**

$x + y = 8,\ 8x + 3y = 39 \Rightarrow x = 3,\ y = 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Compound inequalities

In the system $\{2x + py = 6; qx - 6y = 12$, there are infinitely many solutions. What is $p + q$?

a. -1

b. 0

c. **1 ✓**

d. 2

Notice the constant terms: $12 = 2 \times 6$. Therefore the second equation is twice the first: $q = 2(2) = 4$ and $-6 = 2p \Rightarrow p = -3$. Thus $p + q = -3 + 4 = 1$.

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30 [Intermediate] • Target: Multi-step word problems

For what value of the constant c does the system of equations $2x - 5y = 8$ and $6x - 15y = c$ have infinitely many solutions?

a. 16

b. 20

c. **24** ✓

d. 28

Multiplying the first equation by 3 gives $6x - 15y = 24$. For the system to have infinitely many solutions, the two equations must be identical, so $c = 24$.

31 [Intermediate] • Target: A well-designed word problem: ticket sales at a community theater

In the system of equations: $\{4x + 6y = 15; 2x + ky = 7.5$ If the system has infinitely many solutions, what is the value of k?

a. 2

b. **3** ✓

c. 4

d. 6

Multiplying the second equation by 2 gives $4x + 2ky = 15$. For this to match $4x + 6y = 15$, $2k = 6 \Rightarrow k = 3$.

32 [Intermediate] • Target: Solving linear equations

What is the slope of the line through $(4, 2)$ and $(6, -2)$?

a. -1/2

b. 2

c. -1

d. **-2** ✓

Applying Heart of Algebra principles, $m = \frac{-2-(2)}{6-(4)} = -2$ Therefore, the verified correct option is -2.

33 [Intermediate] • Target: Linear inequalities

In the system $\{mx + 3y = 9; 4x + 6y = 18$, for what value of m does the system have infinitely many solutions?

a. 1

b. **2** ✓

c. 3

d. 4

Multiply first equation by 2: $2mx + 6y = 18$. Matching with $4x + 6y = 18$ requires $2m = 4 \Rightarrow m = 2$.

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34 [Intermediate] • Target: Slope and equations of lines

A particle's position in the xy-plane at time t is given by the parametric equations $x(t) = t^2 - 6t$ and $y(t) = t^3 - 9t$, where $t \geq 0$. The particle starts at the origin and moves along a path. Find the time $t > 0$ when the particle returns to the x-axis, and determine the x-coordinate of this position.

a. $t = 3$**, x-coordinate = -9** ✓

b. $t = 4$, x-coordinate = -8

c. $t = 5$, x-coordinate = -5

d. $t = 6$, x-coordinate = 0

The particle returns to the x-axis when its y-coordinate is 0. We are given $y(t) = t^3 - 9t$. Setting $y(t) = 0$ and factoring, we get: $t^3 - 9t = 0$ $t(t^2 - 9) = 0$ $t(t - 3)(t + 3) = 0$ Since the time $t \geq 0$, the possible values for t are $t = 0$ and $t = 3$. The problem states the particle starts at the origin, which corresponds to $t = 0$ (since $x(0) = 0^2 - 6(0) = 0$ and $y(0) = 0^3 - 9(0) = 0$). We are looking for a time $t > 0$ when the particle returns to the x-axis. This occurs at $t = 3$. Now, we find the x-coordin.

35 [Intermediate] • Target: Systems of equations by substitution

What is the sum of all solutions to the equation $|2x + 2| = 10$?

a. 2

b. 4

c. -6

d. -2 ✓

The equation $|2x + 2| = 10$ splits into two linear cases: Case 1: $2x + 2 = 10 \Rightarrow 2x = 8 \Rightarrow x = 4$. Case 2: $2x + 2 = -10 \Rightarrow 2x = -12 \Rightarrow x = -6$. Sum of solutions: $4 + (-6) = -2$.

36 [Intermediate] • Target: Systems of equations by elimination

Which ordered pair (x, y) satisfies the system of inequalities below? $y, > 3x + 1; y, \leq -x + 10$

a. $(0, -1)$

b. $(1, 6)$ ✓

c. $(6, 16)$

d. $(1, 1)$

Test the ordered pair $(1, 6)$ in both inequalities: 1) $6 > 3(1) + 1 = 4$, which is $6 > 4$ (True). 2) $6 \leq -(1) + 10 = 9$, which is $6 \leq 9$ (True). Both conditions are satisfied.

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37 [Intermediate] • Target: Absolute value equations

If $|3x - 5| = |x + 7|$, what is the product of all real solutions for x?

a. -3 ✓

b. 3

c. -6

d. 6

Case 1: $3x - 5 = x + 7 \Rightarrow 2x = 12 \Rightarrow x = 6$. Case 2: $3x - 5 = -(x + 7) \Rightarrow 4x = -2 \Rightarrow x = -1/2$. Product $= 6 \times (-1/2) = -3$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 5.01 Statistics & Probability

What is the slope of the line given by the equation $4x - 2y = 10$?

a. 2 ✓

b. -2

c. 4

d. $\frac{1}{2}$

Applying Heart of Algebra principles, Solving algebraically gives 2. Therefore, the verified correct option is 2.

39 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 6.01 Geometry

A line passes through the points $(2, 5)$ and $(6, 13)$. What is the slope of the line?

a. 2 ✓

b. 3

c. $\frac{1}{2}$

d. 4

Applying Heart of Algebra principles, Solving algebraically gives 2. Therefore, the verified correct option is 2.

40 [Challenging] • Target: Compound inequalities • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the solution to the equation $2(x + 4) = 5(x - 2)$?

a. 6 ✓

b. 4

c. 8

d. 5

Applying Heart of Algebra principles, Solving algebraically gives 6. Therefore, the verified correct option is 6.

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41 [Challenging] • Target: Multi-step word problems • Cross-Lesson: 8.01 Exponentials & Logarithms

For what value of a does the system $ax + 4y = 7$ and $3x + 2y = 5$ have no solutions?

a. 6 ✓

b. -6

c. 3

d. 4

Lines must be parallel: $\frac{a}{3} = \frac{4}{2} \Rightarrow \frac{a}{3} = 2 \Rightarrow a = 6$.

42 [Challenging] • Target: A well-designed word problem: ticket sales at a community theater • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the sum of all solutions to $|5x - 7| = 10$?

a. **14/5** ✓

b. -3/5

c. 17/5

d. 24/5

$5x - 7 = 10$ or $5x - 7 = -10 \Rightarrow x = 17/5$ or $x = -3/5,\ \text{sum} = 14/5$

43 [Challenging] • Target: Solving linear equations • Cross-Lesson: 3.01 Functions & Graphs

Solve the system of equations: $x + y = 14$ and $x - y = 6$. What is x?

a. 10 ✓

b. 4

c. 8

d. 12

Solving algebraically gives 10.

44 [Challenging] • Target: Linear inequalities • Cross-Lesson: 4.01 Ratios, Rates & Data

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at the point $(-2, 5)$ and the parabola passes through the point $(0, 9)$. What is the value of b?

a. **4** ✓

b. 2

c. -4

d. -2

The vertex form of a quadratic function is given by $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. In this case, the vertex is $(-2, 5)$, so $h = -2$ and $k = 5$. Thus, the function can be written as $f(x) = a(x - (-2))^2 + 5$, which simplifies to $f(x) = a(x + 2)^2 + 5$. The parabola also passes through the point $(0, 9)$. We can substitute these values into the equation to solve for a: $9 = a(0 + 2)^2 + 5$ $9 = a(2)^2 + 5$ $9 = 4a + 5$ $4 = 4a$ $a = 1$ Now we have the function in vertex form.

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45 [Challenging] • Target: Slope and equations of lines • Cross-Lesson: 5.01 Statistics & Probability

How many real solutions does the equation $|2x - 6| = x - 5$ have?

a. **0** ✓

b. 1

c. 2

d. Infinitely many

The absolute value $|2x - 6| \geq 0$ for all real x. For $x - 5 \geq 0 \Rightarrow x \geq 5$. Case 1: $2x - 6 = x - 5 \Rightarrow x = 1$ (not ≥ 5). Case 2: $2x - 6 = -(x - 5) = -x + 5 \Rightarrow 3x = 11 \Rightarrow x = 11/3 < 5$. Neither candidate satisfies $x \geq 5$, so there are 0 real solutions.

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